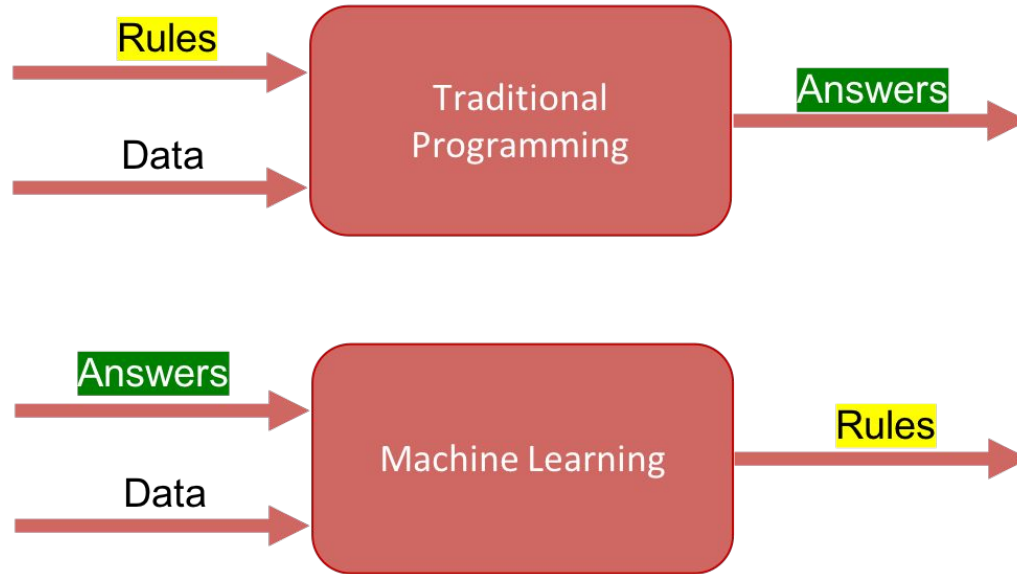


Intro To ML

Course Code - CSE424

Traditional Learning vs Machine Learning



Our Dataset

Input → CGPA

Output → Good / Average Students

Traditional Machine

If the CGPA is lower than a **threshold**, then the output is **average student**.

Else, then the output is **good student**.

threshold = rule

Training Phase Starts

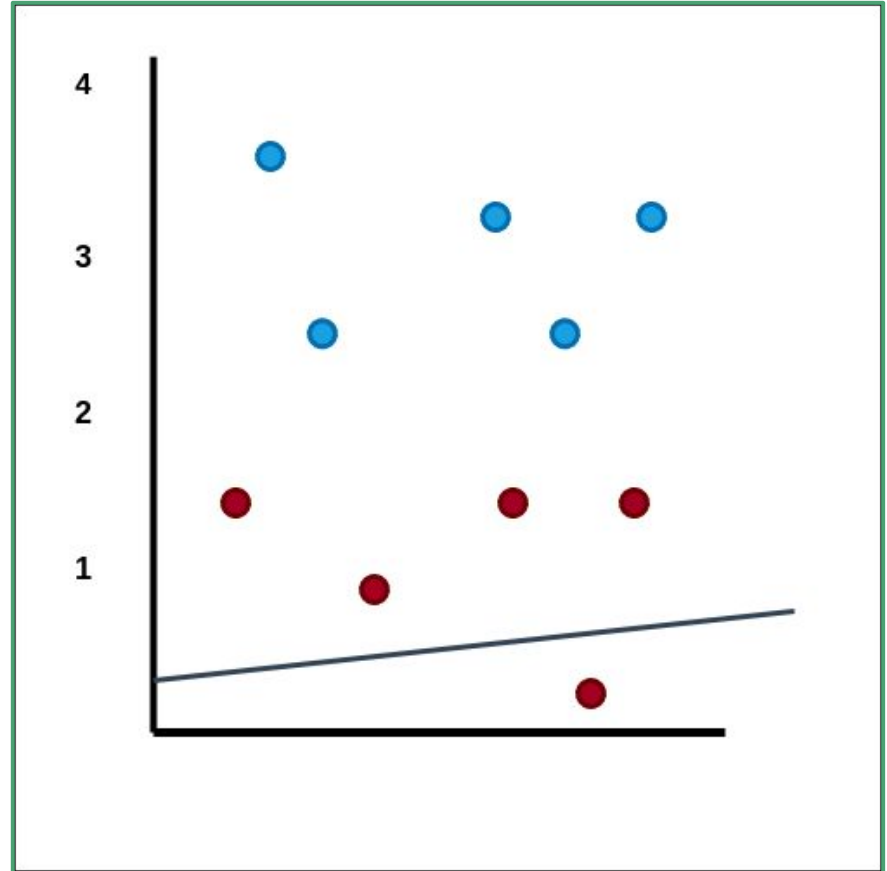
Goal - The line (is the rule) which separates the two labels (good and average) students

Blue - Good Students

Red - Average Students

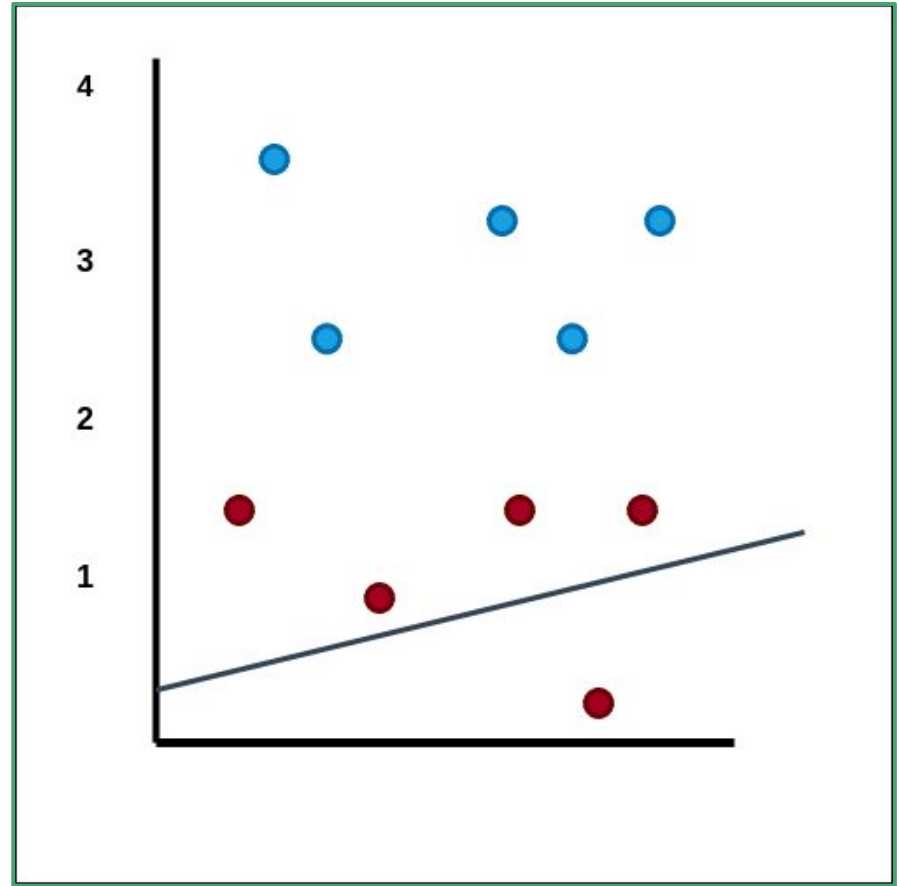
Formulae of a straight line:

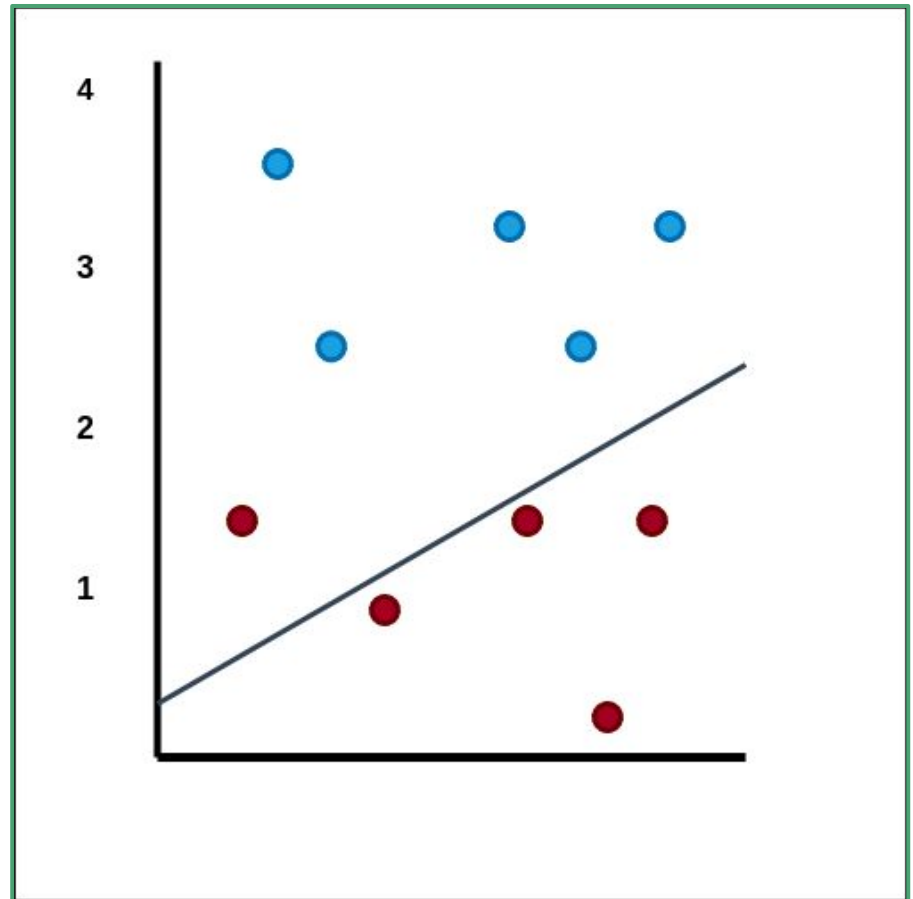
$$Y = mx + b .$$

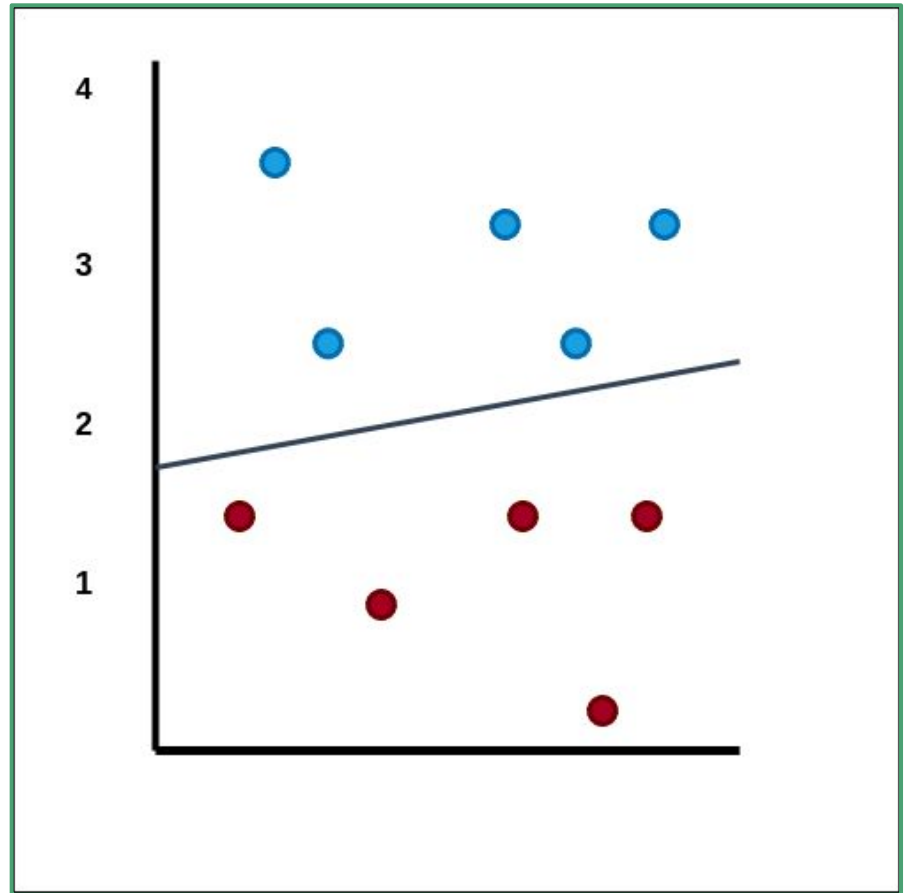


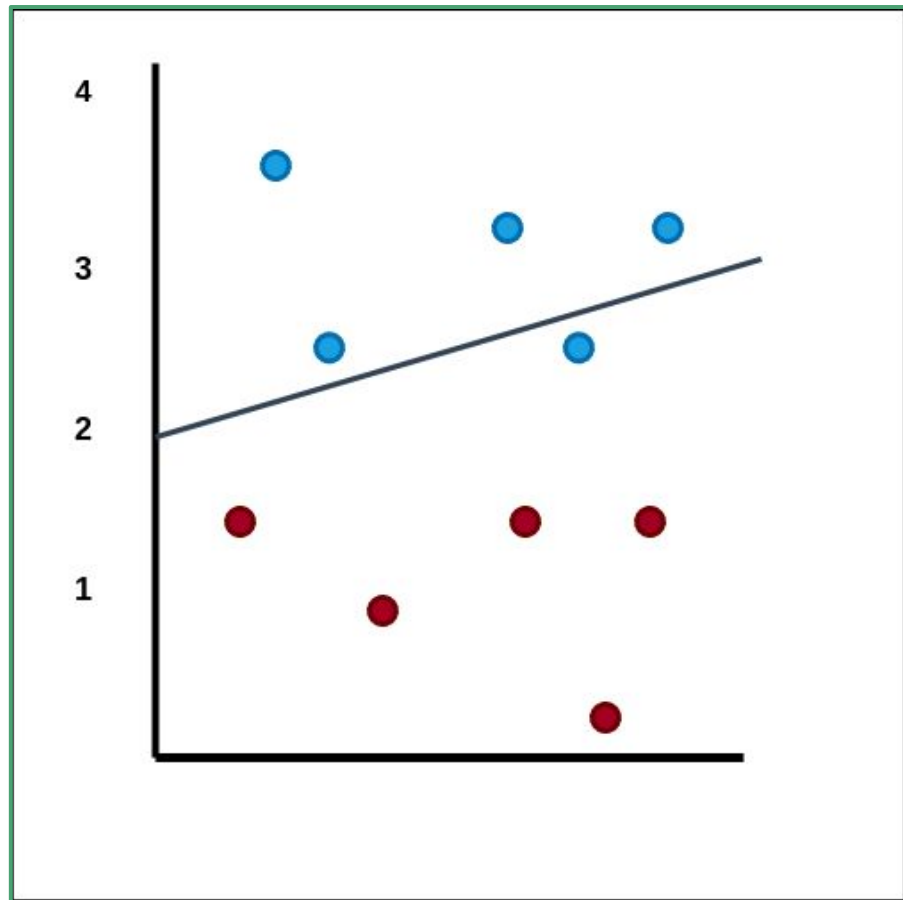
m slightly changed.

Count the number of errors



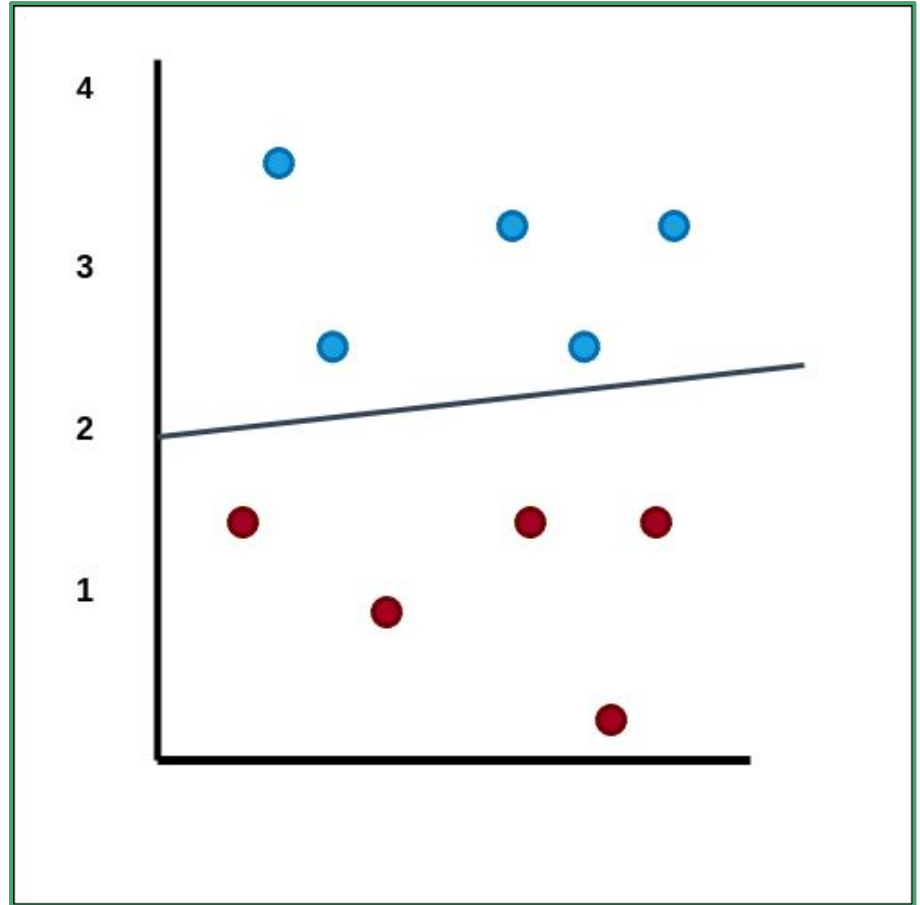






Here, value of m and c changed from the initial value
so that our line splits the
different labels into two
different groups.

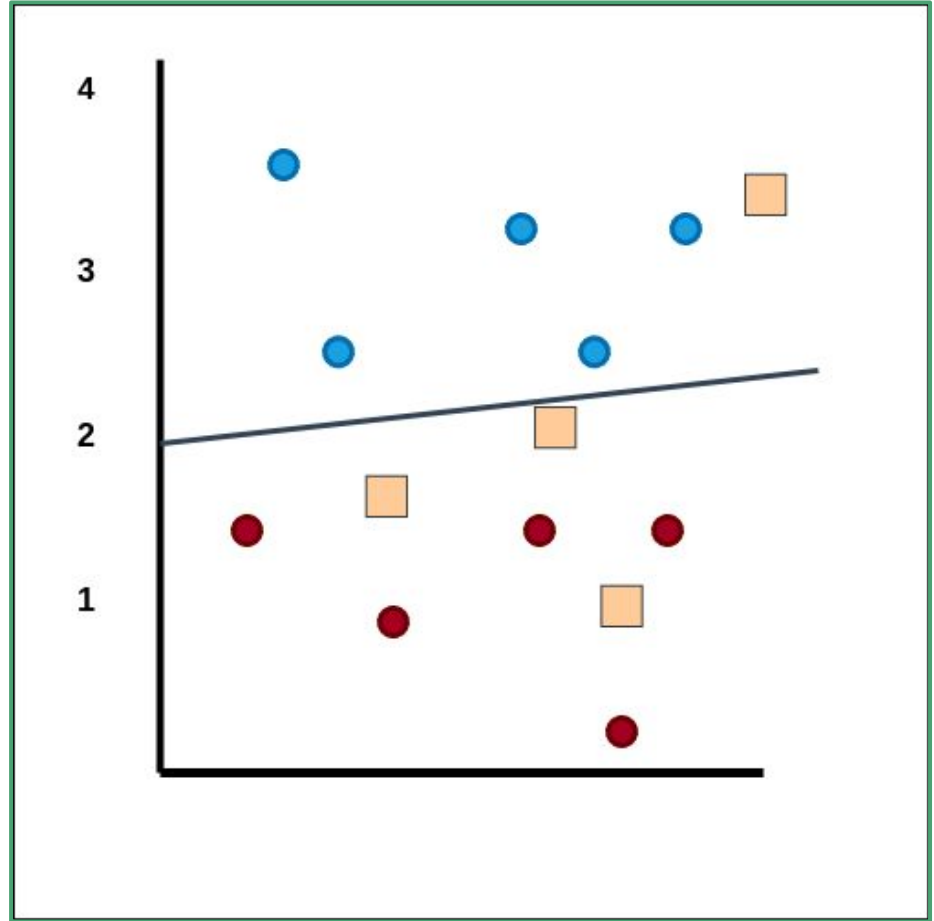
Training COMPLETE



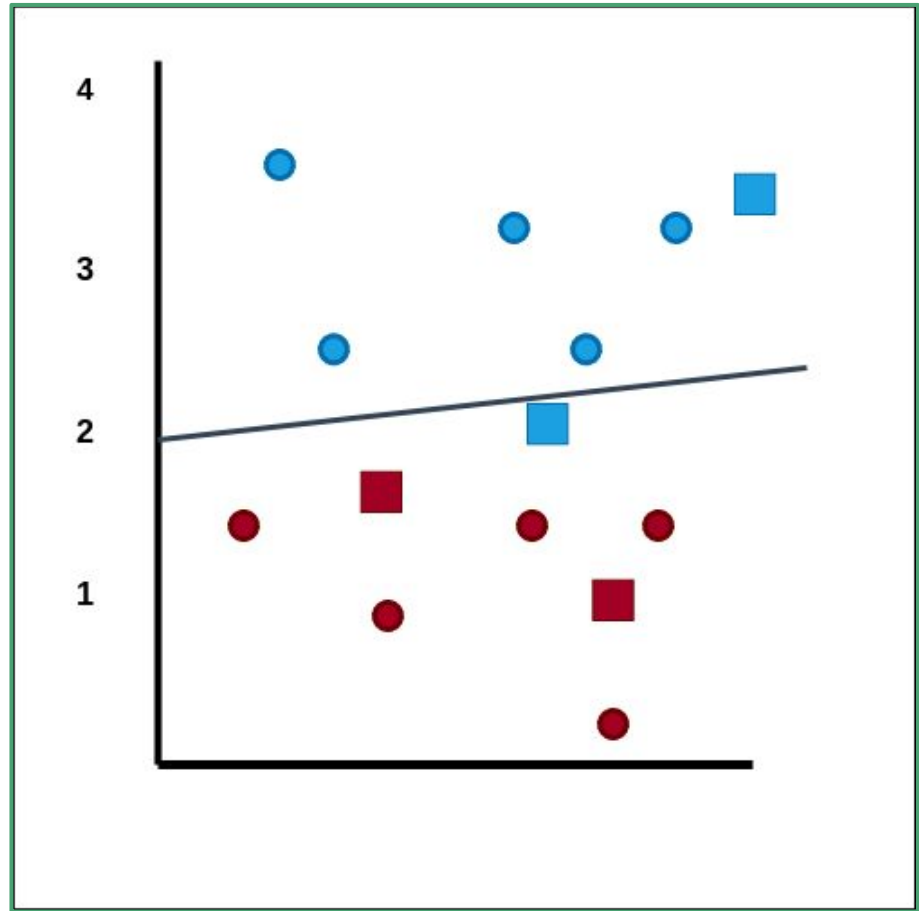
Testing Phase Begins

Above the line - **Blue**

Below the line - **Red**



Calculate the accuracy.



Why do we need preprocessing

- Raw data has many issues such as null values.
- Textual Data that machines can not understand
- Not scaled data

After PREPROCESSING, we feed this data to the model to train - test

Train vs Test Phase

- Train the model to learn the parameters
- Test with unseen data to measure performance of the model

Types of machine Learning

- Supervised Learning
- Unsupervised Learning

Supervised Learning

- Classification
- Regression

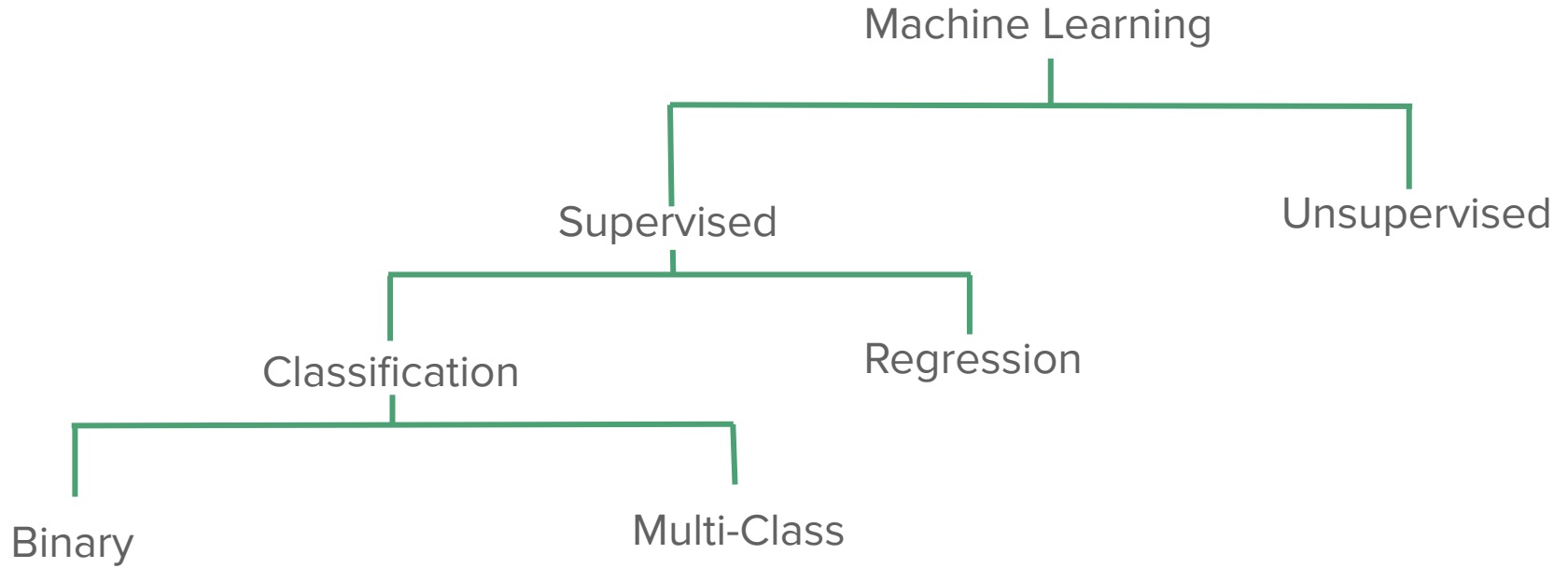
Classification

- Binary (Two classes).

Eg - True or False, Yes or No.

- Multi Classification (More than two classes).

Eg - Best, Average or Bad. Large, Medium or Small.

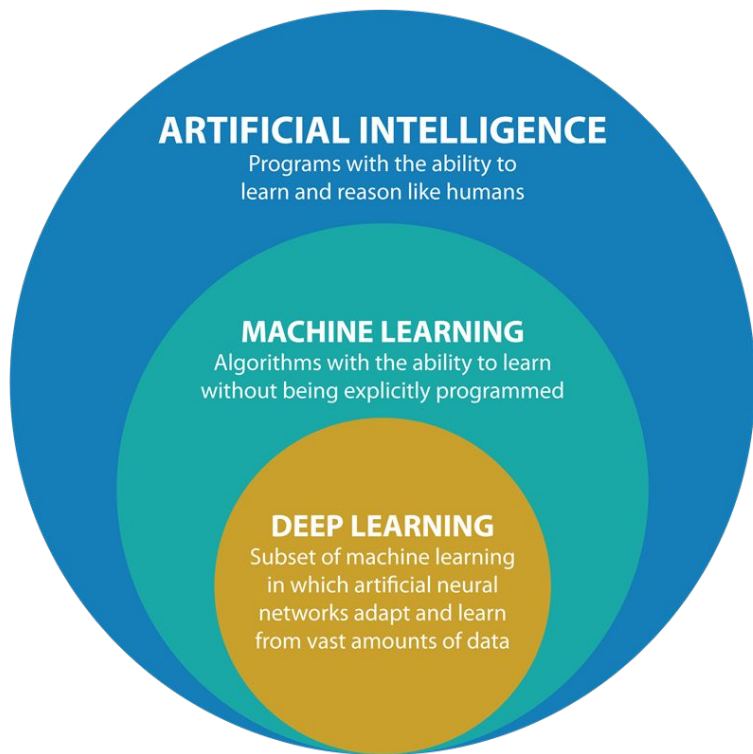


Input vs Output

- Input - Feature Variables
- Output - Target variables / Target Labels

Different types of dataset

- Tabular Dataset
- Image (computer vision)
- Text or speech related (NLP)



- Artificial Intelligence (AI) - CSE422
- Machine Learning (ML) - CSE427
- Deep Learning (DL) - CSE425

Other types of Machine Learning

- Semi supervised Learning
- Reinforcement Learning

ML Pipeline

1. **Data Collection:** Gathering raw data from various sources such as databases, APIs, or sensors.
2. **Data Preprocessing:** Cleaning, transforming, and preparing data for modeling. This involves handling missing values, feature extraction, encoding categorical variables, normalization , etc.
3. **Feature Engineering:** Creating new features or selecting the most relevant features to improve model performance.
4. **Data Splitting:** Splitting the dataset into train and test set. *{Optional} Additionally a validation / dev can be created to perform model tuning.*
5. **Model Training:** Selecting a machine learning algorithm and training it on the processed train set.
6. **Model Prediction:** Using the trained model on a test set to make prediction.
7. **Model Evaluation:** Assessing the trained model using metrics like accuracy, precision, recall, or F1-score on validation data.
8. **Model Tuning:** Optimizing hyperparameters and repeating from **step 5** to improve model performance.

ML - Classification Models

- Logistic Regression
- Support Vector Machine (SVM)
- Naive Bayes
- K-Nearest Neighbors (KNN)
- Decision Tree
- Ensemble Learning (Bagging Classifier, Boosting & Stacking)
-

The End